



Robotic Skin

Researchers at Yale University have developed a Robotic Skin that can turn everyday objects into robots. <https://digit.in/oct18-90-1>



Lion that roars poetries

There's a Lion statue in London's Trafalgar Square that takes words as input and recites AI generated poetries. <https://digit.in/oct18-90-2>

JWST finds evidence of life on two planets in the TRAPPIST-1 System

NASA's James Webb Space Telescope has found biosignatures in the atmospheres of Trappist-1 e and f.

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An international team of researchers used the NASA/ESA/CSA James Webb Space Telescope (JWST) to investigate the atmospheres of the seven Earth sized planets in orbit around the ultracool dwarf star Trappist-1. According to the results, there is a statistically significant indication that biological life forms are modifying the atmospheres of Trappist-1 e and Trappist-1 f. Both the planets are in orbit within the habitable zone of the star, a region where liquid water can exist on the surface.

The findings of the investigation have been published on Wednesday in the journal *Nature Astronomy*, and announced at a press briefing at the NASA Headquarters in Washington.

The Trappist-1 system is named after the Transiting Planets and Planetesimals Small Telescope (TRAPPIST) in Chile, which discovered two of the planets in the system in February 2016. In February 2017, NASA announced the discovery of five additional planets in the system, discovered by the Spitzer space telescope. Follow up observations by the Spitzer and Kepler space telescopes allowed researchers to probe the compositions of the planets in further detail. The observations showed that all the seven planets had a rocky composition, and several could have surface oceans, with more water than the oceans on Earth. Additionally, it was found that the atmospheres of the planets d, e and f did not have high amounts of hydrogen. If the atmospheres had a high hydrogen content, they would not be habitable to organic life.

The Trappist-1 system is located at a distance of about 40 light years from the Earth, towards the constellation of Aquarius. In cosmic distances, the system is relatively close, and can be considered

to be in the stellar neighbourhood of the Earth. As the planets are in another solar system, they are called exoplanets.

Dr Kazumu Umeta, lead author of the study says, "For thousands of years, those who looked into the night skies have wondered if there were aliens on other worlds, gazing towards the Earth. Thanks to the James Webb Space Telescope, we are finally at a point in time where we can begin to answer these questions with certainty."

The findings were made at just about the limit of the capabilities of the JWST. The proximity of the Trappist-1 system, and the brightness of the planets both contributed towards the gathering of the data required. The findings are based on over three years of spectroscopic analysis of the light from the planets. The light is analysed by separating the light into individual wavelengths, and studying these wavelengths of the signatures of chemical components. Traditional approaches for detecting life by studying the atmosphere have not worked very well in the case of the Trappist-1 system. The JWST has previously detected methane in the atmospheres of the Trappist-1 planets, but only trace amounts of oxygen, a requisite for life as we know it on Earth. It was necessary to investigate the geological processes that could account for the methane in the atmosphere. Umeta's team used a new approach called "atmospheric chemical disequilibrium", which measures the composition of the atmosphere, and accounts for the gases through known geological sources. If there is a large imbalance in the expected gases, which cannot be explained by other sources, then the finding can be considered as a biosignature – evidence that there are life forms on the planet.

"The biochemistry of oxygen production is pretty complex. We do not know how rare or common it is in the universe, as we so far have only Earth to go by. Even on this day, we have anaerobic life forms

